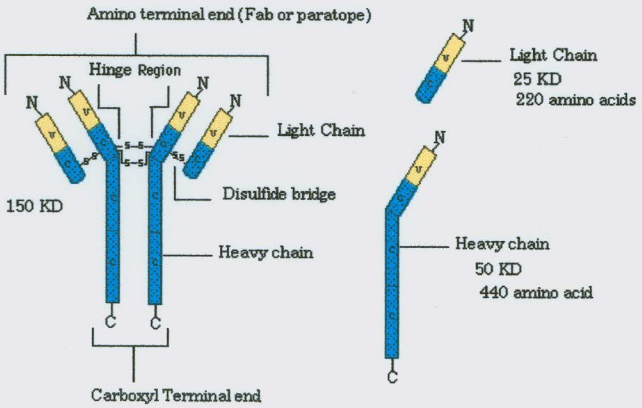
# Biochemistry Cume

April 21, 2006

The schematic drawing of an immunoglobulin is shown below.




What will the expected molecular mass be when you use the following methods?

a. 10pts Molecular mass by gel filtration:
b. 10pts Molecular mass by SDS-PAGE:
c. 10 pts Molecular mass by SDS-PAGE with 2-mercaptoethanol:

Papain cleaves heavy chains in the hinge region, just above the two disulfide bridges that connect the heavy chains, resulting in Fab and Fc fragments.

What will be the expected molecular mass for the papain cleaved fragments when you use the following methods?

d. 10 pts Molecular mass by gel filtration:
e. 10 pts Molecular mass by SDS-PAGE:
f. 10 pts Molecular mass by SDS-PAGE with DTT:

g. 10pts Come up with a method to separate Fab from Fc fragments.
h. 10pts There is no NMR structure deposited in the Protein Data Bank for Fab or intact immunoglobulin. Why not?
i. 10pts Most X-ray structures deposited in the Protein Data Bank are of Fabs. Why?
j. 10pts Describe a use of immunoglobulins in biochemistry research.